



Faculty of System and Process Engineering
M.Sc. Chemical and Energy Engineering

COURSE REPORT — SUSTAINABILITY ASSESSMENT

Carbon Capture In Steam Methane Reforming(SMR)

Life Cycle Assessment (LCA) — Gate-to-Gate Analysis

Methane to SMR process without exploration · CO₂ Mitigation Pathways

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1. Introduction

Carbon capture plays a crucial role in reducing the concentration of carbon dioxide released into the atmosphere by various industries. One significant contributor to carbon dioxide emissions is the hydrogen production industry, particularly through the Steam Methane Reforming (SMR) process. SMR is a conventional method widely used not only for hydrogen production but also in applications such as ammonia synthesis, oil refining, methanol production, chemical manufacturing, fuel cell systems, and other emerging low-carbon energy technologies. This process requires high temperatures to convert methane into hydrogen and carbon dioxide, making it a key focus area for carbon capture efforts aimed at mitigating environmental impact [1].

Steam Methane Reforming reaction: $\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3\text{H}_2$

The products of the reaction are further processed to produce carbon dioxide and hydrogen

Water-Gas Shift (WGS) reaction: $\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$

The carbon dioxide produced during the SMR process is captured before being released into the atmosphere, thereby reducing the carbon content in the air. Separated carbon dioxide, extracted from process gas or flue gas, can be dried, compressed, and prepared for storage or utilization. Therefore, this study focuses on assessing the environmental performance of CO_2 capture in an SMR-based hydrogen production process [2].

2. Goal and Scope

The purpose of this study is to analyze the life cycle assessment (LCA) of the carbon capture system integrated with the Steam Methane Reforming (SMR) process. Specifically, the study aims to evaluate the capture of 1 kg of CO_2 and to assess the effectiveness of this carbon capture system in reducing carbon emissions from SMR-based hydrogen production [3].

This study aims to:

Quantify the material and energy inputs required to capture 1 kg of CO_2 [4, 5]. Identify key areas of environmental impact within the carbon capture system. Evaluate potential improvements in CO_2 separation processes, electricity usage, and CO_2 compression. Facilitate better decision-making to effectively reduce emissions from traditional SMR hydrogen production. The results will provide a foundational understanding of the envi-

ronmental performance of CO₂ capture in SMR and serve as a reference for comparison with other carbon capture or emission reduction technologies.

3. Functional Unit

The functional unit is defined as **1 kg of concentrated CO₂ obtained through the Steam Methane Reforming (SMR) method** [4, 5]. The analysis serves as a foundation for assessing all of the energy and material inputs needed to capture one kilogram of CO₂ produced during SMR in the carbon capture process. It is intended to make direct study of the SMR process with the carbon capture step and to comprehend the decrease in the rate of CO₂ emission rate attributable to the integration of the capture unit [6].

This functional unit is consistent with those adopted in comparable LCA studies of post-combustion carbon capture systems and allows direct benchmarking against the well-documented emissions profile of conventional SMR [4, 7].

4. System Boundary

The boundary is based on the **Gate-to-Gate** method [7]. The natural gas (CH₄) intake to the SMR plant is not explored fully for OpenLCA. It includes the material and energy balance of CH₄ and water with hydrogen gas generation, CO₂, CO slip, and any other energy inputs related to it [8, 4]. The primary focus is on CO₂ absorption and separation from the primary process stream, consumption of steel of the carbon capture unit and the energy input required for capturing the carbon, as well as the compression of CO₂ up to the gate in contrast to the production of hydrogen [5].

The following are excluded from the system boundary [6, 7].

Infrastructure according to plant design and SMR process equipment decommissioning at the end of its useful life are not considered. The transportation, storage and use of the captured carbon are also excluded. The use of co-produced hydrogen gas at the end falls outside the defined boundary.

5. Process Description

5.1 Steam Methane Reforming (SMR).

Steam methane reforming is the dominant industrial route for hydrogen (H_2) production, converting methane (CH_4) and steam (H_2O) into a hydrogen-rich syngas through a catalytic reaction [9, 10]. Methane (CH_4) and steam (H_2O) are blended and preheated to the reaction activation temperature in a feed preheater [9, 11]. The preheated mixture is then fed to the primary reformer, where the reforming reaction takes place. The reaction is endothermic, requiring continuous external heating of the reformer. The reformer outlet stream is a syngas mixture containing hydrogen (H_2), carbon monoxide (CO), unreacted methane (CH_4), and residual steam (H_2O) [9, 12].

5.2 Water-Gas Shift (WGS).

The reformer effluent is cooled in a process-gas cooler before entering the water-gas shift reactor, where additional hydrogen (H_2) is generated by reacting carbon monoxide (CO) with steam (H_2O) [11, 13]. The reactor effluent contains hydrogen (H_2), carbon dioxide (CO_2), unreacted methane (CH_4), residual water (H_2O), and small amounts of unconverted carbon monoxide (CO) [13, 14]. The stream is cooled further to condense the residual water (H_2O), which is removed in a gas-liquid separator [10]. The remaining dry gas advances to a hydrogen-separation unit, where hydrogen (H_2) is recovered as the main product, and a carbon-dioxide-rich (CO_2) tail gas is routed to the capture section [12, 15].

5.3 Downstream Processing and CO_2 Compression.

The off-gas from the pressure swing adsorption (PSA) unit is cooled and routed to the gas-liquid (GL) separator, where residual water (H_2O) is removed and the remaining gas phase containing carbon monoxide (CO), carbon dioxide (CO_2), and unreacted methane (CH_4) is sent to the compound separator for further processing [10, 15]. In the compound separator, carbon dioxide (CO_2) is selectively separated from the gas mixture for capture and storage, while the residual gas stream containing carbon monoxide (CO) and methane (CH_4) is recovered for further use [12, 16]. The separated carbon dioxide (CO_2) stream is then compressed in a three-stage compressor with interstage cooling to achieve the supercritical pressure required for transport and final storage, while the remaining off-gas is preheated and recycled as fuel to the reformer to improve overall energy efficiency and making the process isolated such that CO_2 emission is almost null to the environment [17, 18].

Table 5.1: Input and Product(output) Property Table

Input and Product(output) Property Table					
Object	Steam	Methane	Compressed H ₂ (Product)	Compressed CO ₂ (Product)	Unit
Temperature	150	25	40	65.9942	°C
Pressure	1	1	200	110	bar
Mass Flow	30	10	2.10908	10.2199	kg/h
Molar Flow	1.66525	0.623346	0.998548	0.240206	kmol/h
Volumetric Flow	58.2278	15.42	0.140233	0.0369433	m ³ /h
Density (Mixture)	0.515218	0.648508	15.0398	276.639	kg/m ³

Table 5.2: Chemical Reactions in the system

Reaction	Equation
Methane Reforming	$CH_4 + H_2O \rightleftharpoons CO + 3H_2$ & $CO + H_2O \rightleftharpoons CO_2 + H_2$
Water Gas Shift Reaction	$CO + H_2O \rightleftharpoons CO_2 + H_2$

6. Process Simulation and Modelling

The SMR-CCS process was simulated in DWSIM using the Peng-Robinson thermodynamic package. The flowsheet comprises the steam reformer, a water-gas shift reactor, a gas-liquid separator for water removal, and a PSA unit for H₂ recovery. The CO₂-rich tail gas is routed to a cooler-flash separator and a component splitter for CO₂ separation, followed by a multi-stage compression train for storage.

7. Industrial Analogue

This flowsheet mirrors Air Liquide’s CRYOCAP™ H₂ technology, in commercial operation at Port-Jérôme, France since 2015, capturing 100,000 t/yr of CO₂ at over 97% capture rate [19, 20]. The cooler-flash separator and component splitter in this simulation correspond to its cryogenic purification and PSA/membrane stages (TRL 9). Linde’s HISORP CC is a parallel commercial alternative [21].

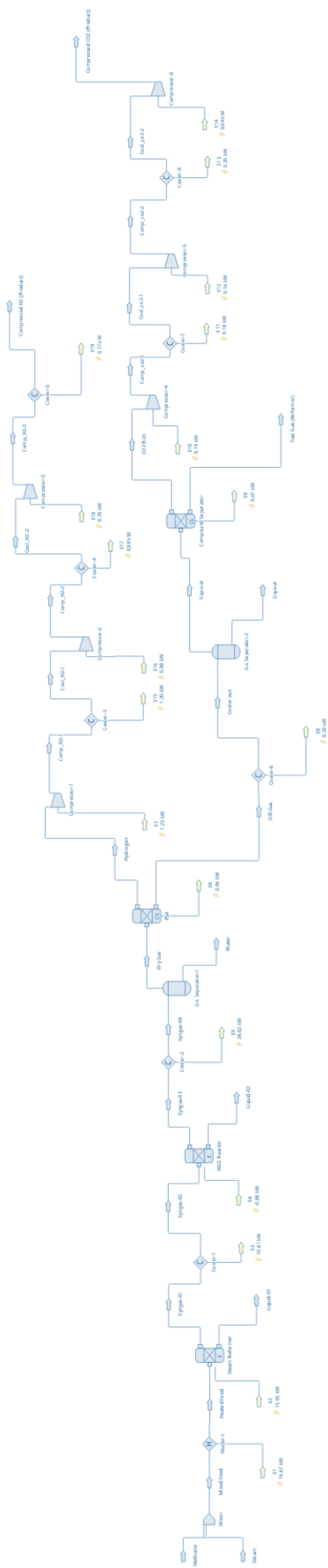


Figure 7.1: Process Flow Diagram of SMR with WGS Reactor, CO₂ compression

Table 7.1: Stream Compositions (Mole Fractions)

Stream	Methane	Carbon Monoxide	Carbon Dioxide	Hydrogen	Water
Cool_H2-2	0.006863	0	0	0.9931	0
Comp_H2-2	0.006863	0	0	0.9931	0
Cool_H2-1	0.006863	0	0	0.9931	0
Comp_H2-1	0.006863	0	0	0.9931	0
Comp_H2-3	0.006863	0	0	0.9931	0
Compressed H ₂ (Product)	0.006863	0	0	0.9931	0
CO ₂ Rich	0.04194	0.006094	0.9474	0.004587	0
Cool_co2-1	0.04194	0.006094	0.9474	0.004587	0
Comp_co2-1	0.04194	0.006094	0.9474	0.004587	0
Cool_co2-2	0.04194	0.006094	0.9474	0.004587	0
Comp_co2-2	0.04194	0.006094	0.9474	0.004587	0
Compressed CO ₂ (Product)	0.04194	0.006094	0.9474	0.004587	0
Off-Gas	0.4589	0.02858	0.3535	0.1506	0.008439
Cooler out	0.4589	0.02858	0.3535	0.1506	0.008439
Vap-out	0.4624	0.02880	0.3561	0.1517	0.0008963
Liq-out	1.201×10^{-8}	2.926×10^{-8}	0.002812	4.107×10^{-7}	0.9972
Fuel Gas (Reformer)	0.6703	0.04002	0.06386	0.2245	0.001339
Hydrogen	0.006863	0	0	0.9931	0
Dry Gas	0.1980	0.01209	0.1495	0.6368	0.003569
Syngas-04	0.1202	0.007337	0.09115	0.3866	0.3947
Liquid-02	0	0	0	0	0
Syngas-03	0.1202	0.007337	0.09115	0.3866	0.3947
Syngas-02	0.1202	0.03672	0.06177	0.3572	0.4240
Liquid-01	0	0	0	0	0
Syngas-01	0.1202	0.03672	0.06177	0.3572	0.4240
Heated Feed	0.2724	0	0	0	0.7276
Mixed Feed	0.2724	0	0	0	0.7276
Steam	0	0	0	0	1
Methane	1	0	0	0	0
Water	2.212×10^{-8}	3.089×10^{-8}	0.001049	4.253×10^{-6}	0.9989

8. Model Graph of OpenLCA

In OpenLCA, a process represents a specific activity or operation within a system, such as chemical absorption, compression, separation, or electricity generation. Each process is characterised by a defined set of inputs (materials and energy) and outputs (products, by-products, emissions, and wastes). By linking multiple processes through their flows, users can construct complete system models in this study, the SMR-based hydrogen production unit with integrated CO₂ capture. OpenLCA assigns environmental impacts to each process using life cycle inventory (LCI) datasets, enabling quantitative assessment of contributions to impact categories such as climate change, fossil resource depletion, and cumulative energy demand. Accurate process definition and consistent flow linking are therefore essential to obtain meaningful and reproducible LCA results.

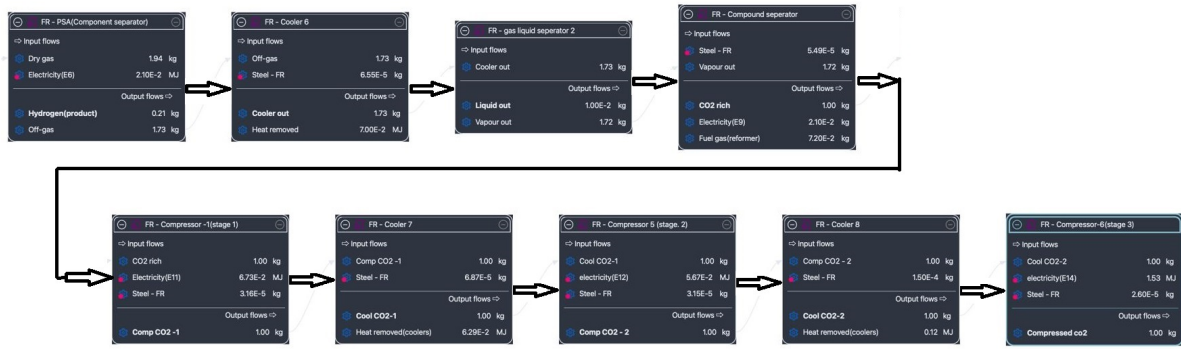


Figure 8.1: OpenLCA process flow diagram for CO₂ compression stages.

9. Impact category result

The following section has the detailed result categories

Table 9.1: Inventory data per kg CO₂ captured for the CO₂ compression system

Component	Steel (kg)	Steel per kg CO ₂ (kg/kg CO ₂)	Electricity (kW)	Electricity per kg CO ₂ (MJ/kg CO ₂)	Heat Removed (kW)	Heat per kg CO ₂ (MJ/kg CO ₂)	Flow Rate (kg/h)	Flow per kg CO ₂ (kg/kg CO ₂)
Cooler 6	105	6.55×10^{-5}	—	—	0.20	7.04×10^{-2}	—	—
Cooler 7	110	6.87×10^{-5}	—	—	0.58	2.04×10^{-1}	—	—
Cooler 8	240	1.50×10^{-4}	—	—	0.35	1.23×10^{-1}	—	—
GL Separator	70	4.37×10^{-5}	—	—	—	—	—	—
Compound Separator	88	5.49×10^{-5}	—	—	—	—	—	—
Compressor 4	50.7	3.16×10^{-5}	0.19	6.73×10^{-2}	—	—	—	—
Compressor 5	50.5	3.15×10^{-5}	0.16	5.67×10^{-2}	—	—	—	—
Compressor 6	41.7	2.60×10^{-5}	0.04	1.53×10^{-2}	—	—	—	—
Methane	—	—	—	—	—	—	10	9.79×10^{-1}
Steam	—	—	—	—	—	—	30	2.94

Table 9.1 presents the inventory data for the CO₂ compression system. All steel values are divided by a scaling factor of 1,635,184 kg, calculated based on a plant lifetime of 20 years, 8,000 operating hours per year, and a CO₂ production rate of 10.2199 kg CO₂/h, to obtain the steel requirement per kg CO₂. Electricity and heat removed values are multiplied by 352.2 MJ/kW, derived by converting the CO₂ mass flow rate from kg/h to kg/s, to obtain the energy requirement per kg CO₂. Methane and steam flow rates (kg/h) are divided by the CO₂ production rate to obtain the mass flow per kg CO₂.

10. Impact Assessment result

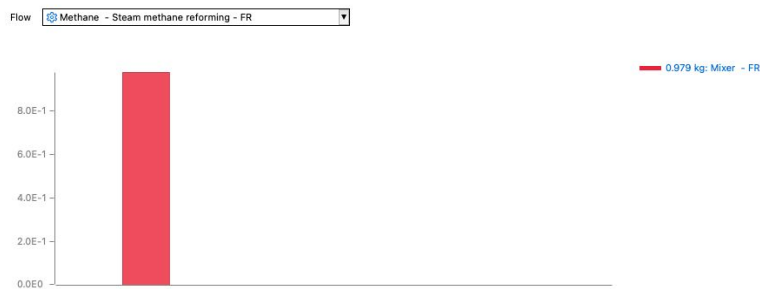


Figure 10.1: Methane required per kg of CO₂ captured.

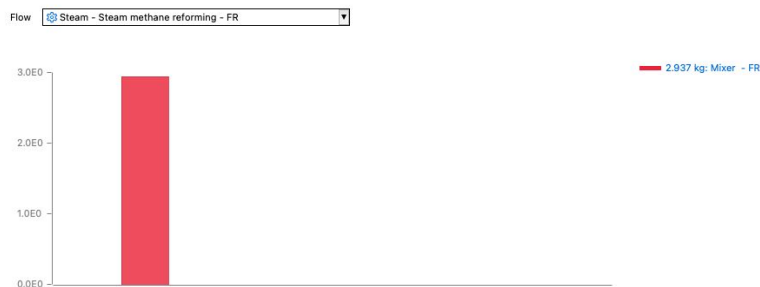


Figure 10.2: Steam required per kg of CO₂ captured.

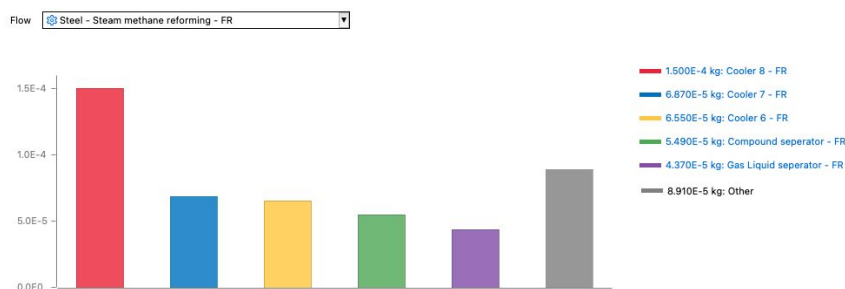


Figure 10.3: Amount of Steel required of each process in carbon capture unit with compare to per kg of CO₂ captured.



Figure 10.4: Energy required for compressor 4 per kg of CO₂ captured.

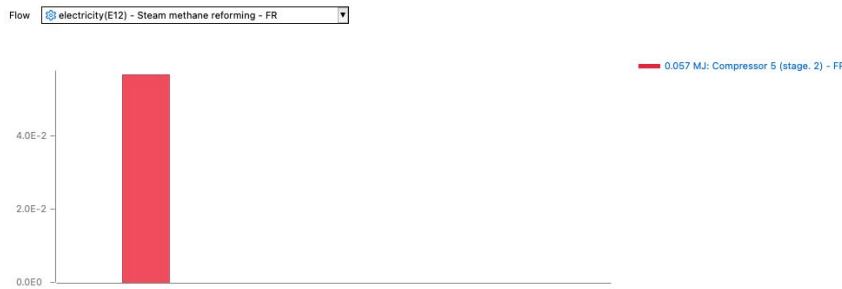


Figure 10.5: Energy required for compressor 5 per kg of CO₂ captured.

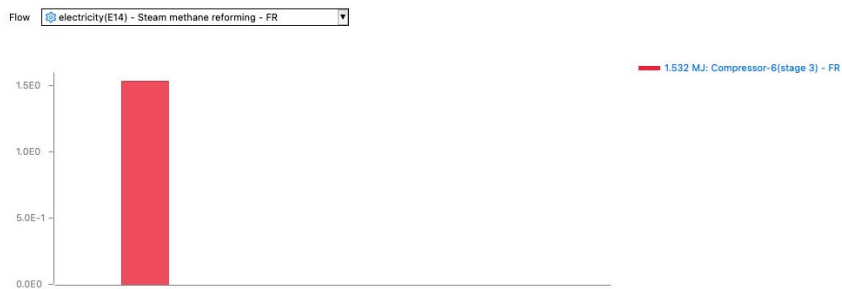


Figure 10.6: Energy required for compressor 6 per kg of CO₂ captured.

11. Conclusion

The DWSIM simulation demonstrates that the carbon capture and compression train requires a total energy input of **1.132 kW** to process the CO₂-rich tail gas from the PSA unit. The off-gas stream entering the capture section contains a CO₂ mass flow of **11.382 kg/h**, of which **10.015 kg/h** is recovered as the compressed CO₂ product stream. This corresponds to a CO₂ **capture efficiency of 88.0%**, indicating that nearly nine-tenths of the carbon dioxide that would otherwise have been emitted to the atmosphere as part of the syngas is successfully captured and made available for storage or utilisation.

The remaining 12% exits with the fuel-gas stream, which is recycled to the reformer furnace as fuel within the plant.

The OpenLCA model for the CO₂ compression and capture system shows that producing 1 kg of captured CO₂ requires approximately 0.979 kg of methane and 2.937 kg of steam as upstream inputs. On the material side, steel consumption per kg of CO₂ captured is dominated by the cooling and separation equipment, with Cooler 8 contributing the largest share (1.50×10^{-4} kg/kg CO₂), followed by Cooler 7 and Cooler 6 (around $6.5\text{--}6.9 \times 10^{-5}$ kg/kg CO₂ each), and smaller contributions from the Compound Separator and GL Separator.

For energy demand, the three-stage compression train shows a sharp increase from Stage 1 to Stage 3: Compressor 4 requires 0.067 MJ/kg CO₂, Compressor 5 requires 0.057 MJ/kg CO₂, while Compressor 6 (Stage 3) requires 1.532 MJ/kg CO₂ – over an order of magnitude higher, due to the higher pressure ratio needed to reach supercritical conditions. This makes the final compression stage and the upstream cooling equipment the two most significant contributors to the overall environmental footprint of the capture system.

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